

Question	Answer	Mark	Mark scheme	Additional guidance
20 (a)	Shown	M1 M1 C1	for $\frac{n}{n+8}$ or starts to work with ratios, eg 3:7 forms equation and clears fractions, eg $10n = 7n + 56$ or $10n + 3(n + 8) = 10(n + 8)$ or equates $\frac{3}{10} = \frac{8}{x}$ or $\frac{3}{10} = \frac{8}{n+8}$ or continues to work with ratios, eg 3:7 = 24:56 gives the total sweets eg $\frac{80}{3}$ oe or number of red sweets $n = \frac{56}{3}$ oe or gives number of red as $\frac{56}{3}$ OR award 3 marks for a complete written argument, eg, $P(y) = \frac{3}{10}$ and there are 8 yellows. This cannot work as 3 is not a factor of 8 (and $\frac{3}{10}$ is in its simplest form)	Does not have to restate the $\frac{7}{10}$; giving a different probability will suffice

Question		Answer	Mark	Mark scheme	Additional guidance
20	(b)	28	P1	for $\frac{n}{n+8}$ and $\frac{n-1}{n+7}$ oe	Note we do not need to see “= 0”; just the LHS is sufficient.
			P1	forms an appropriate equation, eg $\frac{n}{n+8} \times \frac{n-1}{n+7} = \frac{3}{5}$	
			P1	for correctly forming a quadratic ready for solving, eg $an^2 + bn + c (= 0)$, $2n^2 - 50n - 168 (= 0)$, $n^2 - 25n - 84 (= 0)$ oe	
			P1	process to solve quadratic equation, ft a 3 term quadratic factorising eg $(n + 3)(n - 28) (= 0)$ oe or completing the square or correct use of formula eg $\frac{- - 25 \pm \sqrt{25^2 - 4 \times - 84}}{2}$, $\frac{- - 50 \pm \sqrt{50^2 - 4 \times 2 \times - 168}}{2 \times 2}$	
			A1	cao	Award 0 marks for a correct answer with no supportive working.